

Final Interview Challenge

Scenario & Evaluation

You will design and implement a solution and submit an initial prototype one week from the date of the interview. Your work will be evaluated on:

- Coding proficiency
- UI/UX quality
- Creativity
- Ability to interpret requirements
- Ability to deliver working features

You may use any tools, frameworks, or AI solutions. **The solution must include both a Web and a Mobile version**, as our actual project involves supporting both platforms.

Deadline for full implementation: One week from the date of the interview.

Problem Statement: 4D/TOTO Application

Build a simple application that supports Singapore 4D **and** TOTO use cases across **both Web and Mobile platforms** (Android, IOS & Mobile Web Browser View)

Core Features

1. Ticket Upload

Users must be able to take a photo of their 4D/TOTO ticket and upload it through **both the Web and Mobile applications**.

2. Number Extraction (OCR)

Automatically process the uploaded photo to extract:

- Number combinations
- Draw date
- Game type (4D / TOTO)
- Ticket format variations
- System bet category (for TOTO)

Your solution should handle typical ticket layouts and common variations.

3. Data Storage

Store extracted data — including number combinations, purchase date, draw date, game type, and expanded system bet combinations — in a database as part of the user's history.

4. Results Retrieval

Your system should retrieve official winning numbers using **web scraping or API (if any)**. However, retrieval behavior must be context-aware:

Case A — For Future Draws (results not released yet)

If a user uploads a ticket **before the draw date**, the system must:

- Continuously check for official 4D/TOTO results
- Stop once results for that specific draw date have been released
- Proceed to compare results and notify the user

Case B — For Past Draws (results already released)

If a user uploads an **old ticket**, or a ticket dated weeks earlier, the system should:

- Retrieve the **existing past results immediately**
- Compare the ticket numbers without performing continuous polling

This should make the system more efficient and robust.

5. Notifications

Once results are retrieved (immediately or after continuous checks), notify the user:

- Whether their ticket has won
- The prize category / matching tier
- Relevant draw information

Notifications must work on both Web and Mobile.

6. User Purchase History Page (New Requirement)

Your application must include a dedicated page — available on both Web and Mobile — that displays the user's full ticket purchase history. This page should show:

- Uploaded ticket images (or thumbnails)
- Extracted number combinations

- Game type (4D or TOTO)
- Purchase date and draw date
- System bet type (if applicable)
- Winning or non-winning status once results are available
- Prize tier (if any)

The history page should allow:

- Sorting (e.g., newest to oldest, winning first)
- Filtering (e.g., only 4D, only TOTO, only system bets, only winning tickets)
- Clicking into each entry for detailed view (full OCR results, processing status, etc.)

This requirement ensures users can track their previous uploads and results clearly, mirroring real world usage patterns.

7. Past Results Page

Include a dedicated page (Web + Mobile) that automatically retrieves **all past winning numbers** for:

- 4D
- TOTO

Data may be retrieved via web scraping or a suitable public API (if available).

8. Predictive Analysis Extension

In addition to scraping past 4D and TOTO results, your application must also include a dedicated **Prediction Page** (on both Web and Mobile).

This page should:

1. Use past results data collected from the **Past Results Page**.
2. Apply **three distinct predictive analysis models** to generate predictions.
3. For each model, produce:
 - **One predicted 4D number**
 - **One predicted TOTO set**

TOTO Prediction Requirements (Extended)

When generating the TOTO prediction:

- The prediction must follow a **System 12 format**.
- This includes:
 - **6 primary numbers**
 - **Additional supplementary numbers** up to a full System 12 set (i.e., **12 numbers**

total) Documentation Requirements for Predictive Models

For each of the three predictive models used, clearly document:

- Why the model was chosen
- Its core assumptions
- How the model works (methodology)
- Its evaluation or validation approach
- Confidence indicators or uncertainty levels
- A clear disclaimer stating that all predictions are purely for **educational purposes** and **not intended for gambling**

This requirement ensures candidates demonstrate not only technical implementation but also the ability to reason about model selection, limitations, and responsible presentation of predictive output.

TOTO System Bets Requirement

Your application must correctly detect and process TOTO System bets (e.g., System 7, 8, 9, etc.). The system must:

- Identify whether the ticket is Standard or System
- Expand all possible 6-number combinations from the system bet
- Store expanded combinations
- Compare all expanded combinations against official results

User Manual Requirement

You must produce a **highly detailed** User Manual. It will be used as part of the assessment. The manual must include (but is not limited to):

- Full installation and setup instructions
- What applications or dependencies must be installed
- How to run the Web app and Mobile app

- How to configure the environment
- Step-by-step instructions with screenshots
- How to test each feature
- Troubleshooting steps
- Any known issues
- Predictive analysis usage guide
- Data privacy and security considerations

We will assess the User Manual heavily, because future interns must be able to understand your work and continue development using your documentation alone.

Submission Guidelines

Deliverables:

- Source code repository
- Deployable build (Web + Mobile)
- User Manual (PDF/Markdown)
- Optional demo video

Documentation must include:

- README with quick-start guide
- Architecture overview
- API endpoints
- Data source citations
- Ethics statement (no gambling use)

Evaluation Criteria

- Completeness of features (upload, OCR, storage, results retrieval, notifications)
- Correct handling of TOTO System bets
- Code quality and testing
- UI/UX quality
- Accuracy of predictive analysis
- Documentation clarity (especially User Manual)
- Robustness & reliability

Timeline

Prototype: Core user flow working with at least one example ticket Full implementation: One week from interview date

Optional Bonus Features

- Multi-language support
- Accessibility features
- Offline mock API/data for demo purposes
- Export/reporting for ticket history and results